File permissions in Linux

Project description

In this activity, I analyze and update file permissions for an organization's research team to ensure data security, Using Linux command line tools, I audit current access settings and identify configurations that violate company policy, I then modify permissions to restrict unauthorized access, aligning the system with the principle of least privilege

Check file and directory details

To check the permissions of the files and directories, I first navigated to the correct folder using `cd /home/researcher2/projects`. Then, I executed the `ls -la` command. The `ls` command lists the files, the `-l` flag displays them in a long listing format to show the 10 character permissions strings, and the `-a` flag ensures that hidden files (like `.project_x.txt`)

are visible

```
researcher2@4eb512930d65:~$ pwd
/home/researcher2
researcher2@4eb512930d65:~$ cd /home/researcher2/projects
researcher2@4eb512930d65:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Sep 10 13:46 .
drwxr-xr-x 3 researcher2 research_team 4096 Sep 10 14:47 ..
-rw--w---- 1 researcher2 research_team   46 Sep 10 13:46 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Sep 10 13:46 drafts
-rw-rw-rw- 1 researcher2 research_team   46 Sep 10 13:46 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Sep 10 13:46 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Sep 10 13:46 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Sep 10 13:46 project_t.txt
researcher2@4eb512930d65:~/projects$
```

Describe the permissions string

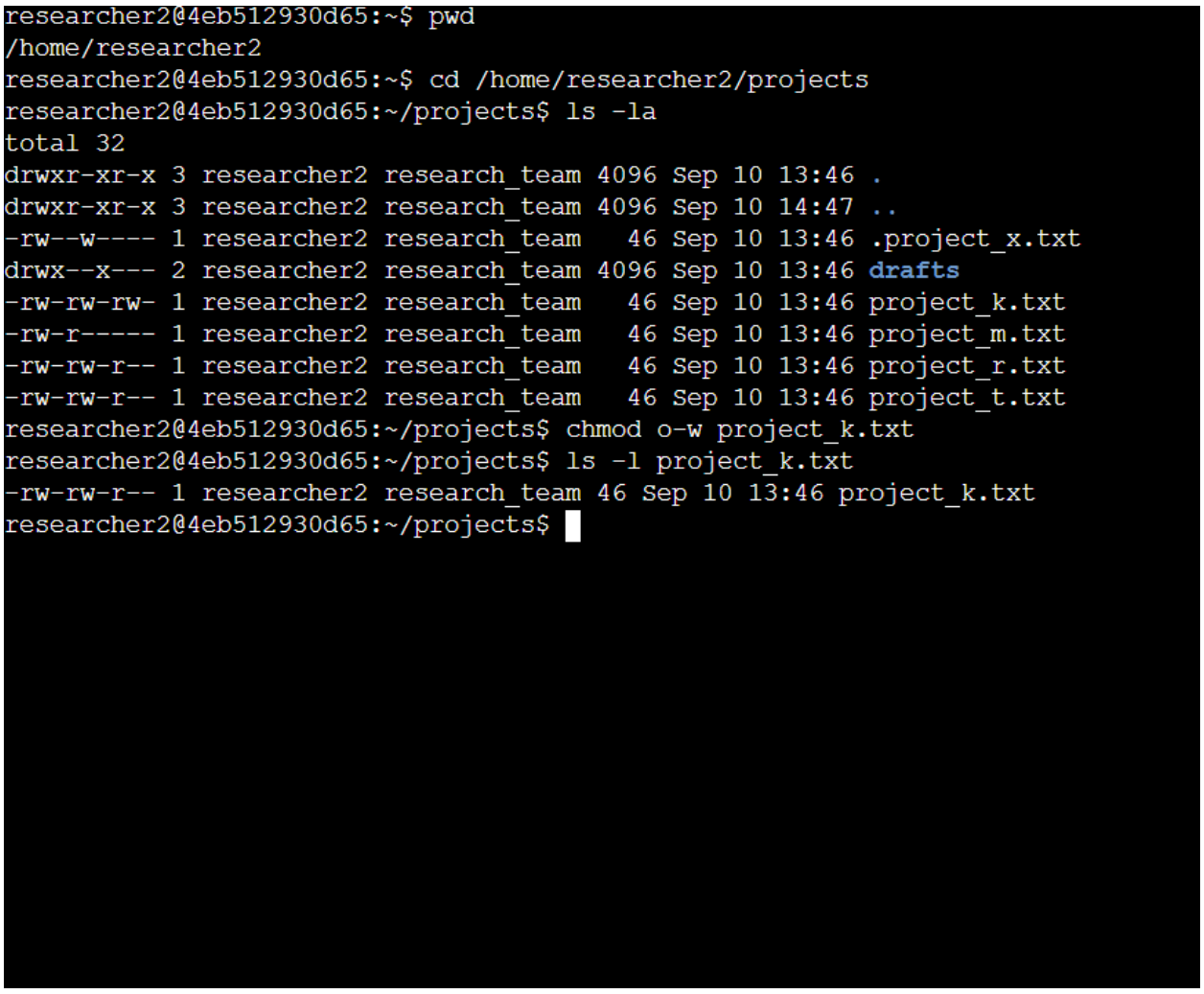
For my example, I selected `project_k.txt`, which has the 10 character permissions string **`-rw-rw-rw-`**. This string breaks down as follows:

- **Position 1 (`-`):** Represents the file type, The hyphen indicates it is a regular data file, not a directory.
- **Positions 2–4 (`rw-`):** Represents **User (owner)** permissions. The owner (`researcher2`) has read (`r`) and write (`w`) access, but not execute (`-`) access.
- **Positions 5–7 (`rw-`):** Represents **Group** permissions. Members of the `research_team` group have read (`r`) and write (`w`) access.

- **Positions 8–10 (`rw-`):** Represents **Others** permissions. Anyone else on the system has read (`r`) and write (`w`) access

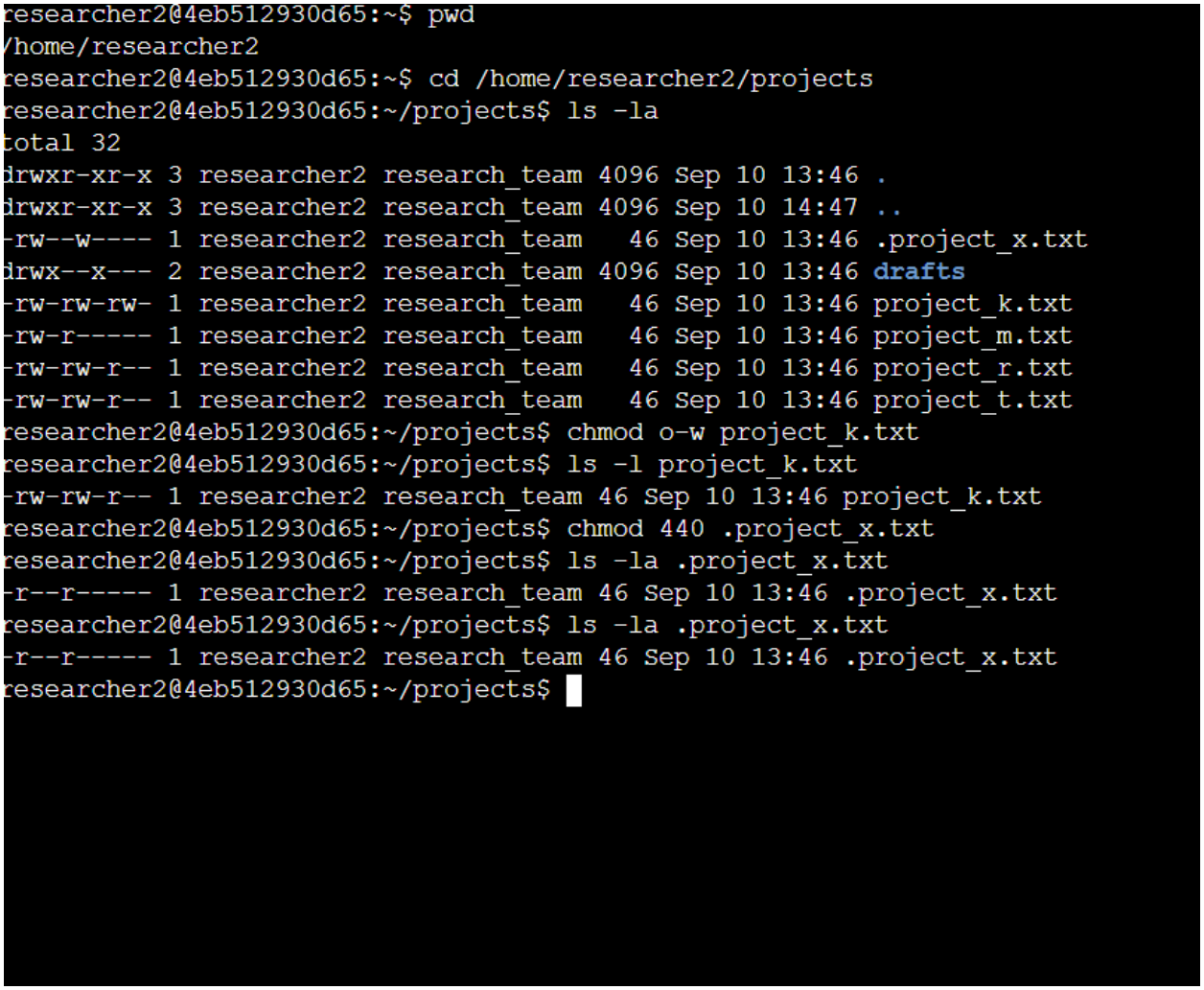
Change file permissions

Based on the initial system audit, the file `project_k.txt` violated organizational security standards because the public "others" group possessed active write permissions (`-rw-rw-rw-`), To resolve this vulnerability and remove write permissions from public users, I executed the command `chmod o-w project_k.txt` Running a long listing verification command confirms the string updated successfully to `-rw-rw-r--`, establishing proper access limitations



Change file permissions on a hidden file

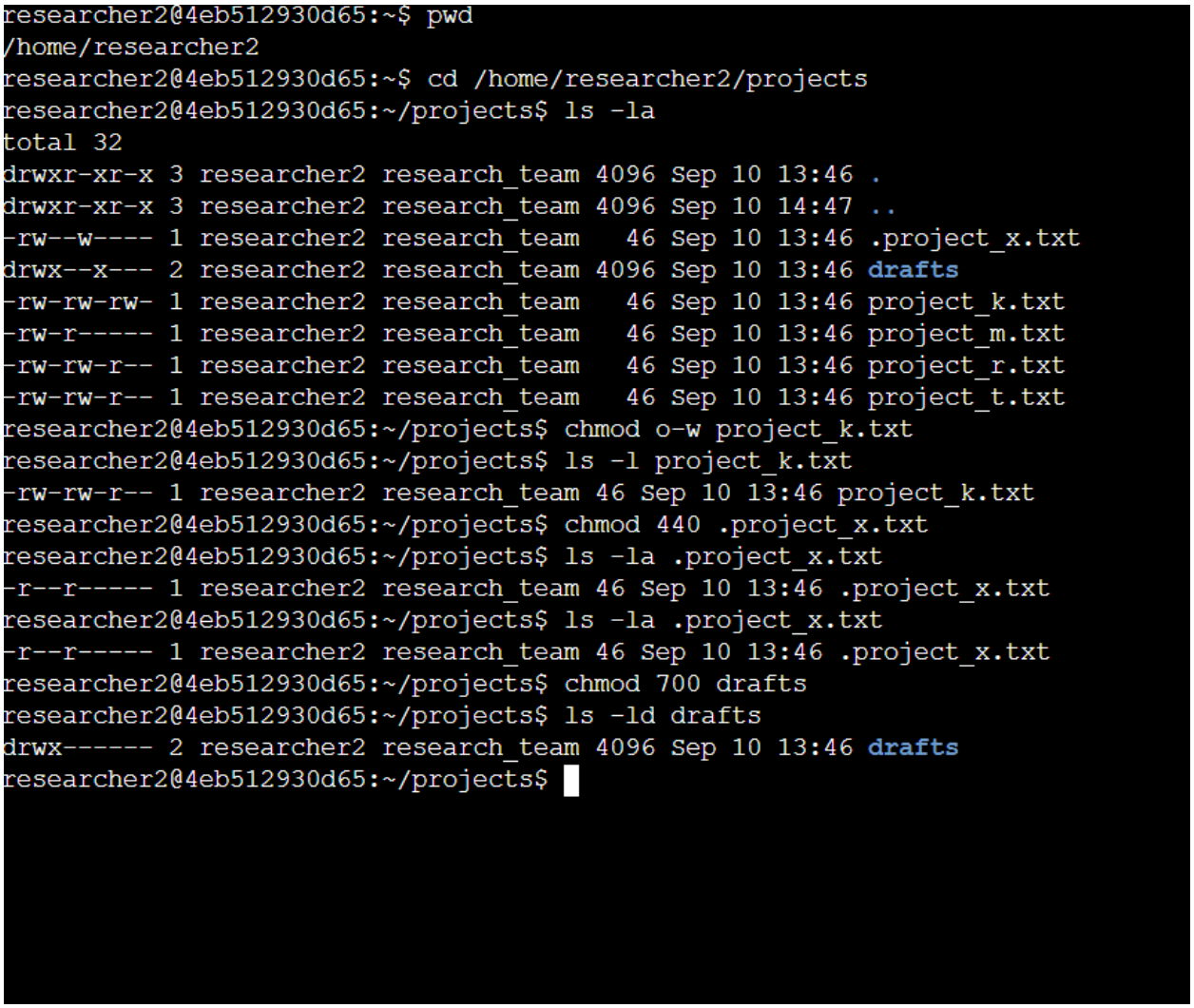
The archived hidden file `.project_x.txt` required specific restrictions to protect its contents from unauthorized modifications while maintaining readability for the team, To remove all write privileges while allowing the owner and group to read it, I ran the command `chmod 440 .project_x.txt`. Verifying the file details with `ls -la` confirms the permissions string successfully shifted to `-r--r-----`, restricting access exactly as required by company compliance guidelines



Change directory permissions

The `drafts` directory contains sensitive operational files that must remain strictly confidential, to keep the integrity and ensure that only the owner `researcher2` can access this directory and its contents while blocking all

access for the group and others, I executed the command `chmod 700 drafts`. Verifying the setup with `ls -ld drafts` confirms the directory permissions updated to `drwx------`, successfully isolating the folder according to the principle of least privilege



Summary

Through this security assignment, I reviewed system access settings using `ls -la` and updated permissions to match company compliance guidelines. Using targeted `chmod` commands, I removed public write access from key project files, restricted access to the sensitive `drafts` directory, and properly isolated hidden archive files. These changes safeguard data

integrity, protect files from unauthorized edits, and show how to enforce basic access controls in a Linux environment